

End to End Encrypted IoT

London IoT Meetup - September 2022



A brief introduction



Engineer, Atsign & Dart GDE
building a platform that puts people in
control of their data.

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@cpswan



Agenda

- Why?
- Overview of Mobile World Congress Demo
- Security, Privacy
- Review



Why?

Your Thing - Your Data

Overview of Mobile World Congress Demo



All the code is open source

atsign-foundation / mwc_demo Public

generated from atsign-company/archetype

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

trunk 8 branches 4 tags

Go to file Add file Code

 cpswan Punctuation bd9e731 13 days ago 131 commits

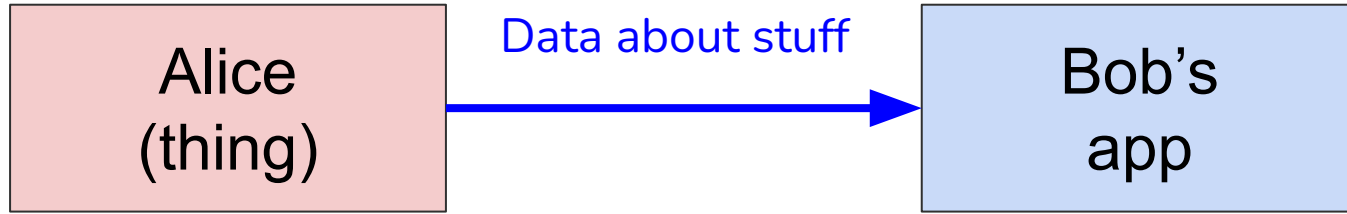
.github	Initial commit	last month
at_cli	Script to read HR from MQTT and publish to @sign	last month
dart/iot_sender	Split out max30101 package into its own repo	14 days ago
flutter/iot_reciever	apple version in store	29 days ago
gsm	add ppp.sh	15 days ago
python	More detailed message	15 days ago
shell	Reduce range	last month
systemd	systemd units	15 days ago
.gitignore	Remove hive files and update .gitignore	last month
CONTRIBUTING.md	correct repo name etc.	15 days ago
LICENSE	Initial commit	last month
README.md	Punctuation	13 days ago
code_of_conduct.md	Initial commit	last month



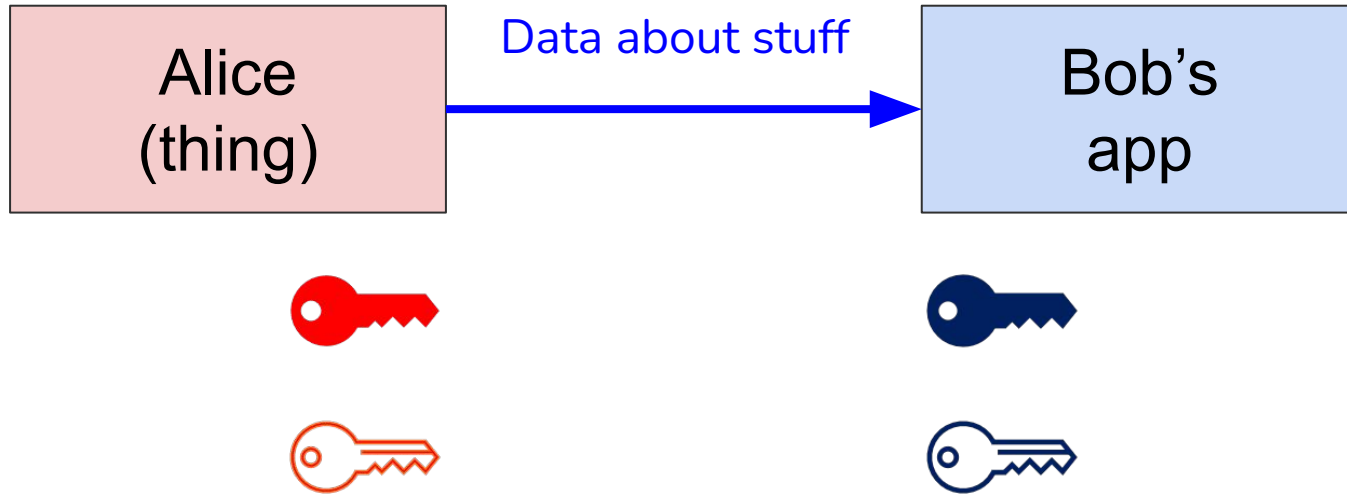
https://github.com/atsign-foundation/mwc_demo

Connectivity, Security, Privacy

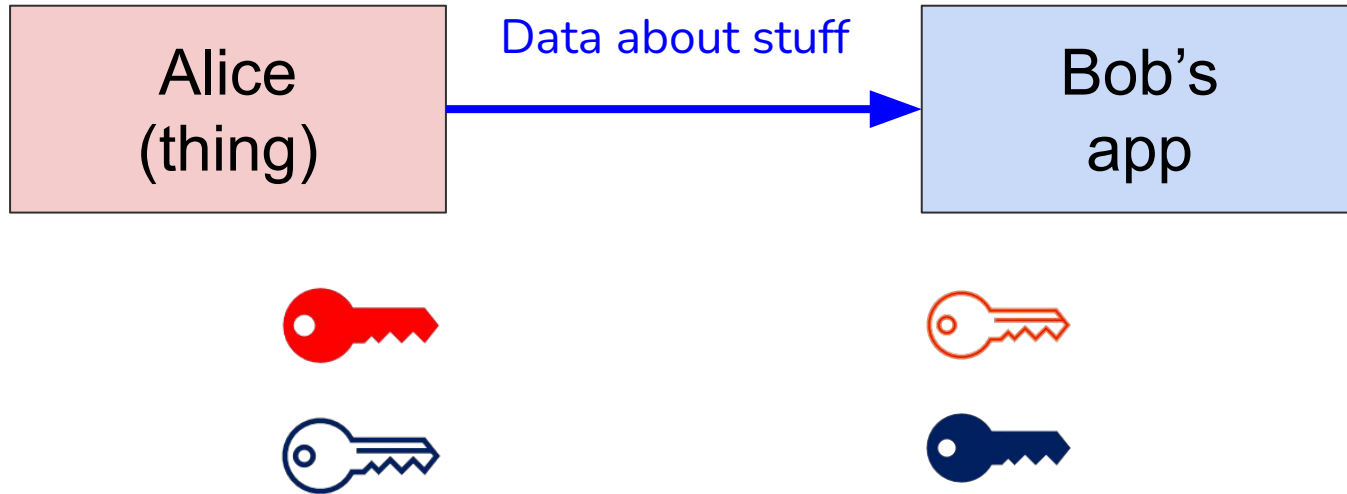
Alice_thing - Bob_person_app



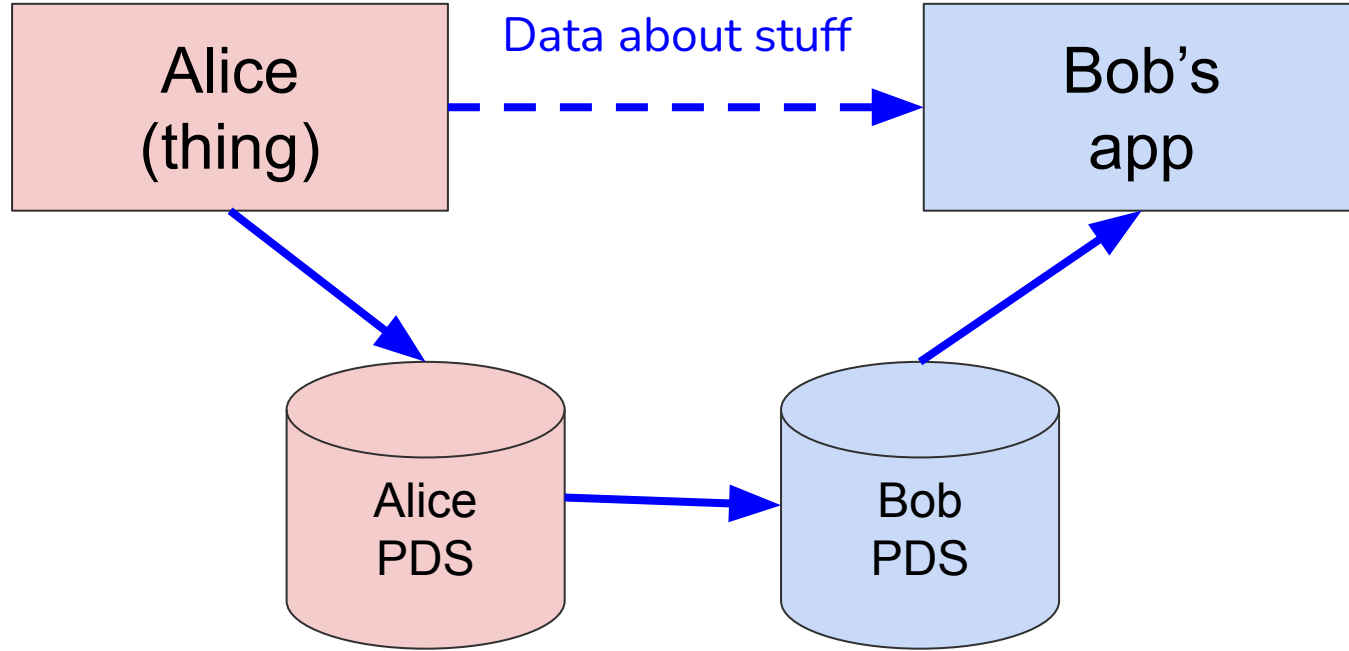
Alice and Bob each have public and private keys



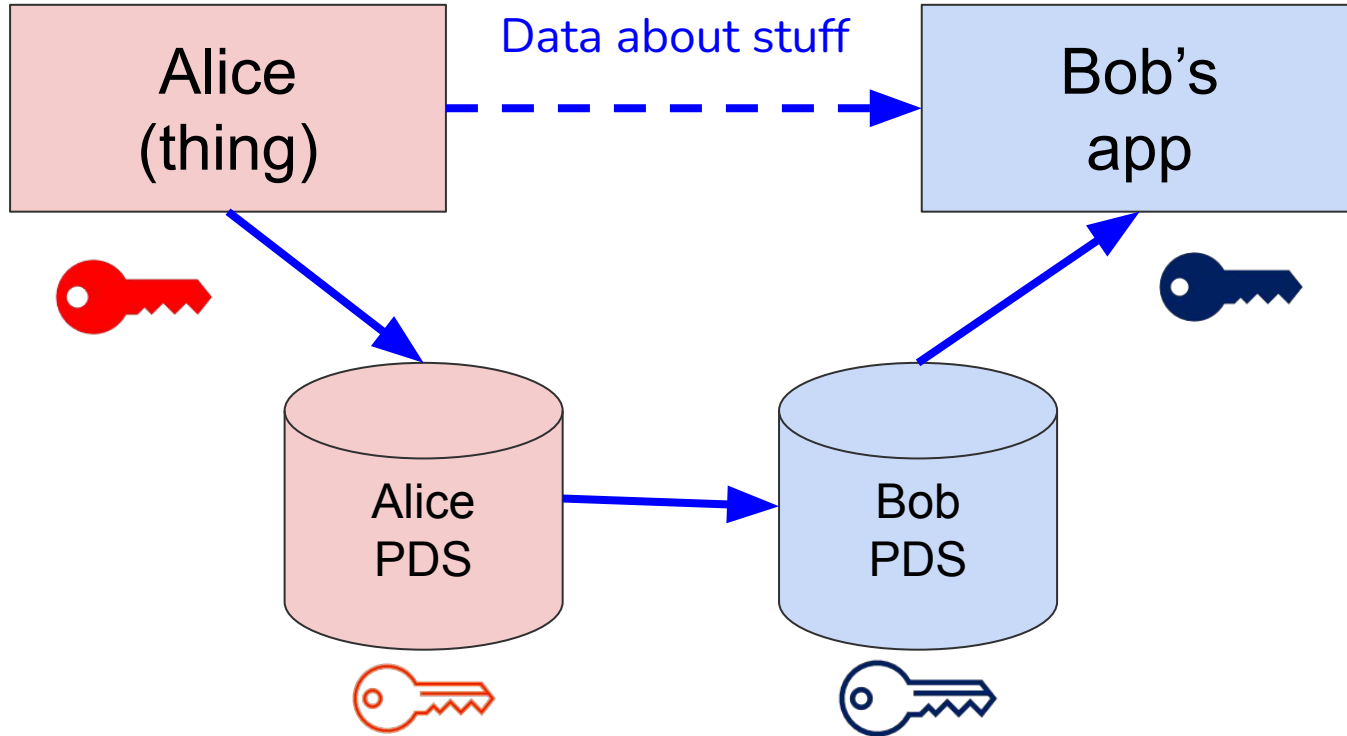
Public keys need to be exchanged



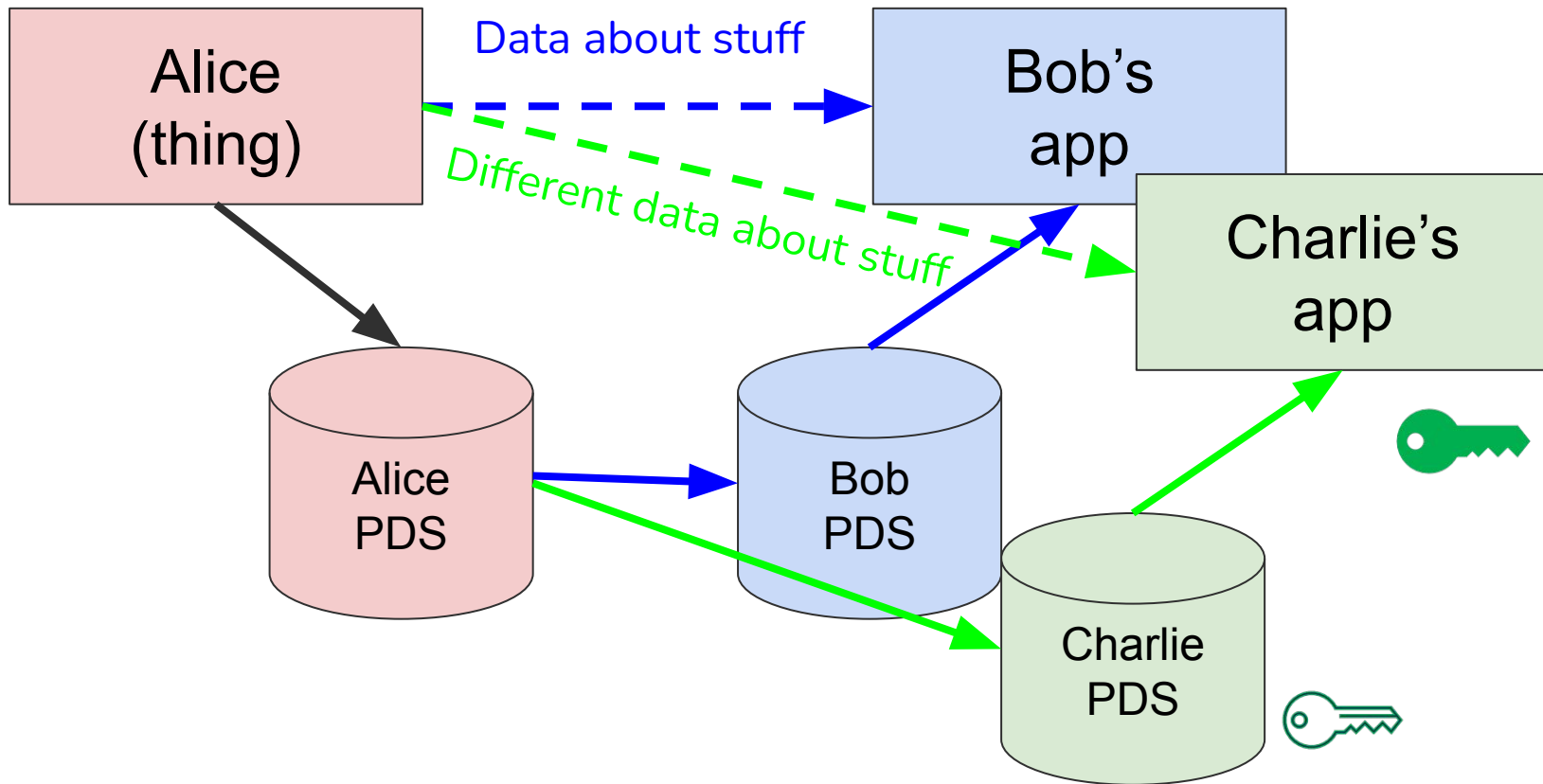
Introduce personal data stores



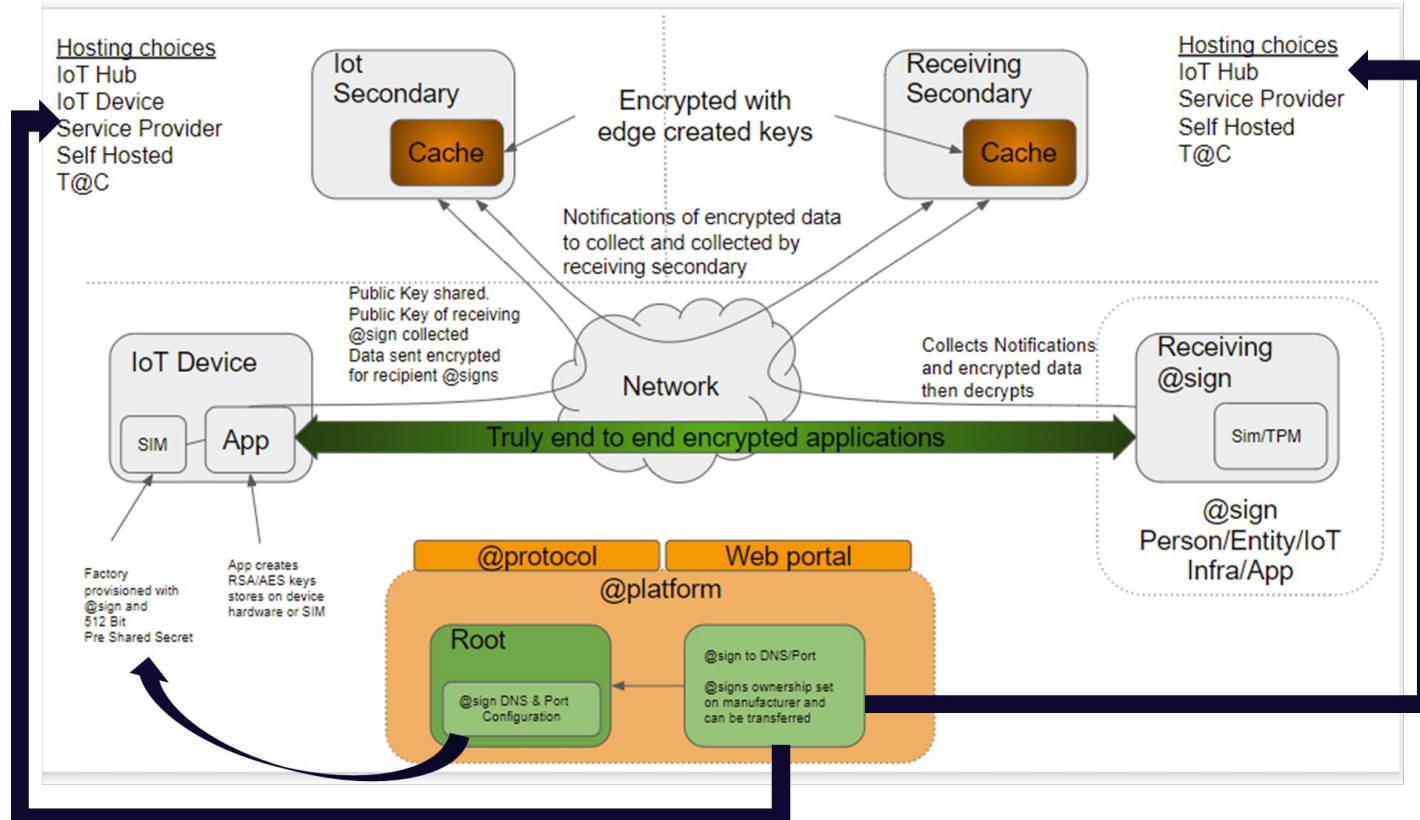
This helps us manage key exchange



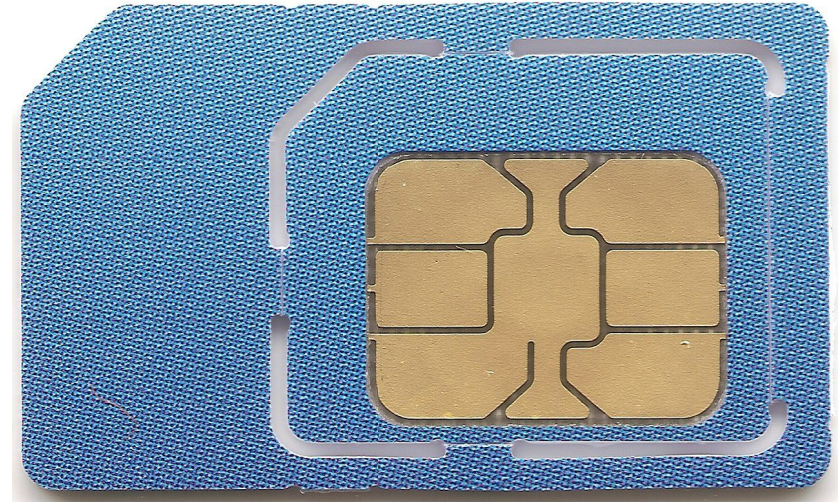
Differential Privacy



Naming services etc. creates a full end-end platform



Where do the keys go?



[Keys by Jay Gooby](#) (CC BY 2.0)



Admin access - Ssh! No ports



Ssh! No ports

ssh no ports provides a way to ssh to a remote linux host/device without that device having any open ports (not even 22) on external interfaces. All network connectivity is out bound and there is no need to know the IP address the device has been given. As long as the device has an IP address, DNS and Internet access, you will be able to connect to it.



<https://github.com/at-sign-foundation/sshnoports>

Review

→ Why?

◆ Your Thing, Your Data

→ Overview of Mobile World Congress Demo

◆ End to end encrypted data from device to app(s) with ability to send different data to different people

→ Security, Privacy

◆  Personal data stores simplify connectivity and key management



Thanks for your time

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Get started with the atPlatform



<https://docs.atsign.com/>



<https://github.com/atsign-foundation>



@atsigncompany

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Questions?